Policy and Stakeholder Analysis Report

Deliverable 1.2

24 February 2025

A transdisciplinary research project with funded by the EU and the UKRI

Pandemic literacy and viral zoonotic spillover risk at the frontline of disease emergence in Southeast Asia to improve pandemic preparedness

PANDASIA

Policy Analysis and Stakeholder Mapping Report

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Project's Partners

No.	Abbreviation	Full name	Country	Main roles
1	CU	Chulalongkorn University	Thailand	WP1 lead
2	MU	Mahidol University	Thailand	WP2 lead
3	KKU	Khon Kaen University	Thailand	WP3 lead
4	UMU	Umeå University	Sweden	WP4 lead
5	UKHD	Universitätsklinikum Heidelberg University	Germany	WP5-lead
6	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
7	NMBU	Norwegian University of Life Sciences	Norway	The Coordinator (WP7)
8	QMUL	Queen Mary University of London	UK	WP1- Partner
9	NVI	Norwegian Veterinary Institute	Norway	WP4- Partner
10	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP2- Partner

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1. Executive Summary

This is an interim report for the WP1's Deliverable 1.2. The introduction of the report provides the contextual background and rationale for the policy document analysis and stakeholder mapping using a one health perspective. This is followed by an in-depth methodology and methods section detailing data selection, collection and analysis processes for the interviews, focus group discussions, co-production workshops and policy documents in relation to guiding frameworks. An overview of the emerging results from the rapid assessment are shared focusing on 12 key themes, and the ongoing full assessment is introduced. The report concludes by discussing the next steps to fulfil task 1.2.

2. Introduction

The PANDASIA project uses a transdisciplinary approach to understand the drivers of potential zoonotic spillover risks in Thailand, with the focus on two diverse provinces, Chiang Rai in the north of Thailand bordering Myanmar and Laos, and Chanthaburi in the east of Thailand bordering Cambodia. WP1 is the social science team that is providing the baseline to set the scene for understanding the context, policies, actors and stakeholders' perspectives, roles, knowledge and attitudes towards the risk and drivers of zoonotic spillover in these two provinces. While our research project is focused on local context, the local context cannot be understood outside wider structures of national, provincial and local policies that shape the actions, knowledge and attitudes of individual and collective actors on community level. In turn, local realities and experiences of interactions, barriers and risks to zoonotic spillover are necessary to be incorporated in national and provincial policies to protect human, animal and environmental health and prevent zoonotic spillovers in the future.

There is a strong body of literature focusing on policies and experiences of preparedness, to prevent the spread of infectious disease (Al-Tawfiq et al., 2024; Kojima et al., 2024; Nicastri et al., 2019; Regmi et al., 2015). Biopolitics and control of peoples and populations behaviour have become essential to pandemic preparedness and responses. Over the last three decades, biosecurity has become an important topic in relation to border security and border control (Dittrich et al., 2023, Hinchliffe et al., 2013). Since the SARS pandemic in 2002-2004, and with the latest COVID-19 pandemic, public health has been politicized in governance of populations, nations and within international relations (Fisher et al., 2021, Ristic et al., 2022, Ullah et al., 2022).

While this downstream policy approach is essential, the upstream policy approach to a healthy interrelationship between environment, animal and human health is of even more importance when we want to prevent the potential spread of an emerging disease in the in the first place (Markotter et al., 2023) . A One Health approach, encompassing the health of animals, human and their environment are guiding international frameworks and national policies to achieve upstream policies for healthy environments and pandemic prevention (WHO et al., 2022).

This is the focus of the policy analysis and stakeholder mapping of PANDASIA. The task addresses RR1 and RR2 from the overall WP1.

RR1. Human and societal factors of importance for zoonotic spillover risk that may impact human, animal, and environmental health identified

RR2. Key policies, policy actors, and stakeholders relevant to zoonotic spillover risk prevention and control measures at local and national levels identified.

This report will provide an overview of the methodology and methods used for the task and insight into the preliminary results and further analysis.

3. Methodology and Methods

WP1 applied a policy and stakeholder mapping methodology that draws upon the Walt and Gilson (1994) policy triangle to analyse context, content and process of policies in relationship to a specific disease and actors. The triangle places actors at the centre, acknowledging actors and stakeholders are those who play an active role and have agency to influence and act upon or are influenced by a policy. Alongside the policy triangle, one-health and eco-health frameworks are used to understand human, environmental and animal health interactions embedded in ecosystem functioning (Charron, 2006, Coker et al., 2011, Desvars-Larrive et al., 2024).

1.1 Data Selection and Group Definition

Three groups of stakeholders were chosen representing the different levels and governmental and non-governmental activities in Thailand. The three groups were defined as below:

Group 1 – Government officers for central, regional, provincial, district and sub-district human and animal, environmental officers and municipalities

Group 2 – Non-governmental NGOs, INGOs and civil societies

Group 3 – Local community members, village health volunteers, animal health workers, teachers and monks

1.2 Data Collection

Key In Depth Interviews (KII)

Key in-depth interviews (KII) were conducted in Chanthaburi in September 2023 with group 1 and group 3 participants, in Chiang Rai in January 2024 with group 3 participants and additional interviews were conducted between August 2024-October 2024 online with group 1 and group 2 participants. All in person interviews were conducted in Thai and online interviews were in English. In Chanthaburi and Chiang Rai the IDIs were conducted in sub-district health centres. Each interview was audio recorded, and notes were taken by the interviewer. Each interview lasted approximately an hour. All interviews were transcribed and when in Thai they were translated using AI with human translation quality checks.

Focus Group Discussion (FGDs)

Focus group discussions (FGDs) were conducted in Chanthaburi in September 2023 with group 3 participants, Bangkok in October 2023 with group 1 and 2 participants and in Chiang Rai in January 2024 with group 3 participants. All FGDs were conducted in Thai. In Chanthaburi and Chiang Rai the FGDs were conducted in sub-district health centres and in Bangkok in a hotel. Each FGD lasted approximately an hour and a half and included on average 6 participants, a facilitator and note taker. All FGDs were transcribed and translated to English using AI with human translation quality checks.

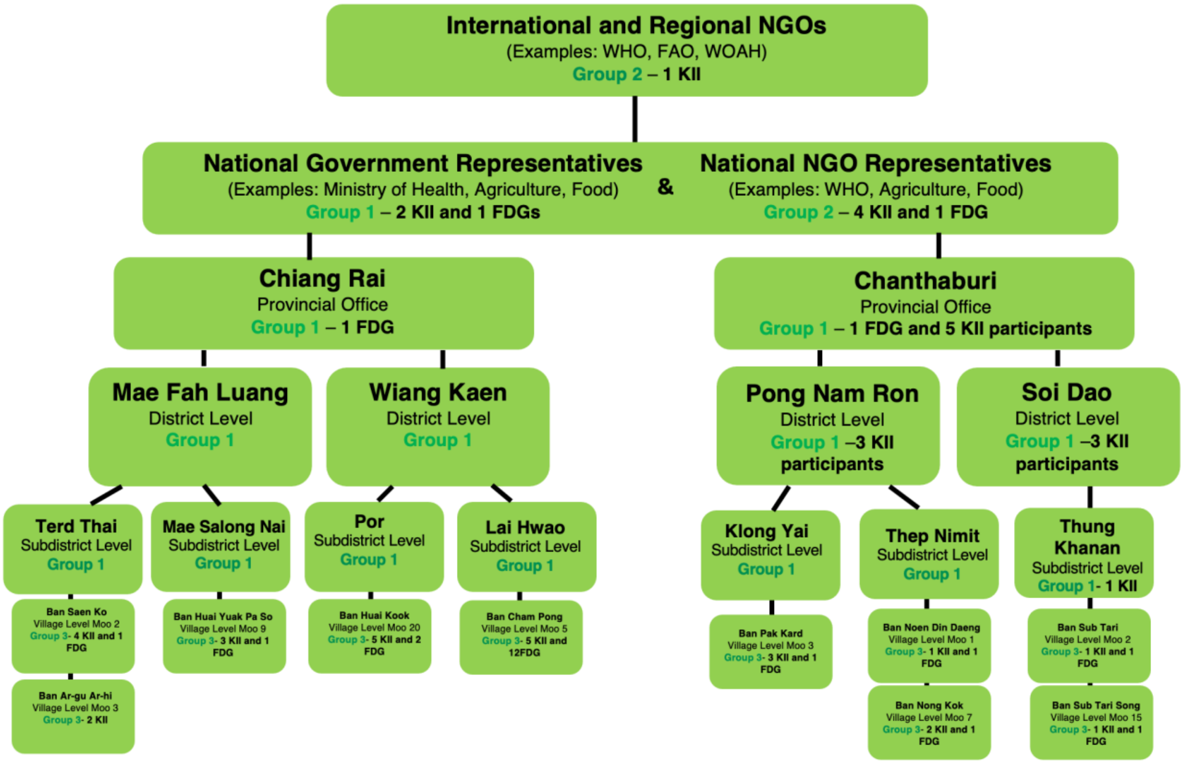

Figure 1: A visualisation of the participants recruited for the policy and stakeholder mapping interviews and focus group discussions.

Co-Production Workshop

The co-production workshops consisted of presentations by participatory institutions and multiple focus group discussion activities with broad questions and scenarios to be addressed in small groups. The workshops were conducted over three days with four diverse groups of actors including group 1 and group 2 (Figure 1). Groups were asked to create mind maps for each activity. Each group was composed of stakeholders, a facilitator and a notetaker. All activities were audio recorded, transcribed and translated using AI and human translator summaries were also provided. Each activity lasted approximately 2 hours.

Policy Document Collection and Selection

The policy document selection process used multiple methods including requesting policy documents from participants, sending official request letters to relevant organisations and conducting online searches for publicly available international and national policies. The selection process was intended to capture all relevant one-health related policies that would inform all the work packages and the overall research project. International, national and local policies were selected to understand how policies are adapted to local contexts. Policies were included based on the following criteria: dated from 2017 onwards, local and provincial policies in Chanthaburi or Chiang Rai, national/bilateral/regional/global policies, policies on one health / forestry or conservation / domestic disease control/ livestock disease control / wildlife disease control / encroachment / land use change/ hunting wildlife / wet market / wildlife trade/ zoonotic disease prevention, control and surveillance/ emerging disease/ street vendor/ market hygiene and sanitation/ water treatment and quality/ migration/ labour/ climate change mitigation and adaption/ biodiversity conservation. Policies were excluded if not in Thai or English, before 2017, related to clinical treatment or not related to potential zoonotic spillover risk.

1.3 Data Types

Multiple data sources were collected. These included interviews, interviewer notes, focus group discussions (co-production workshops were considered as a form of focus group discussions for the purpose of this research), facilitator notes, debriefs, diagrams, translator summaries and policy documents.

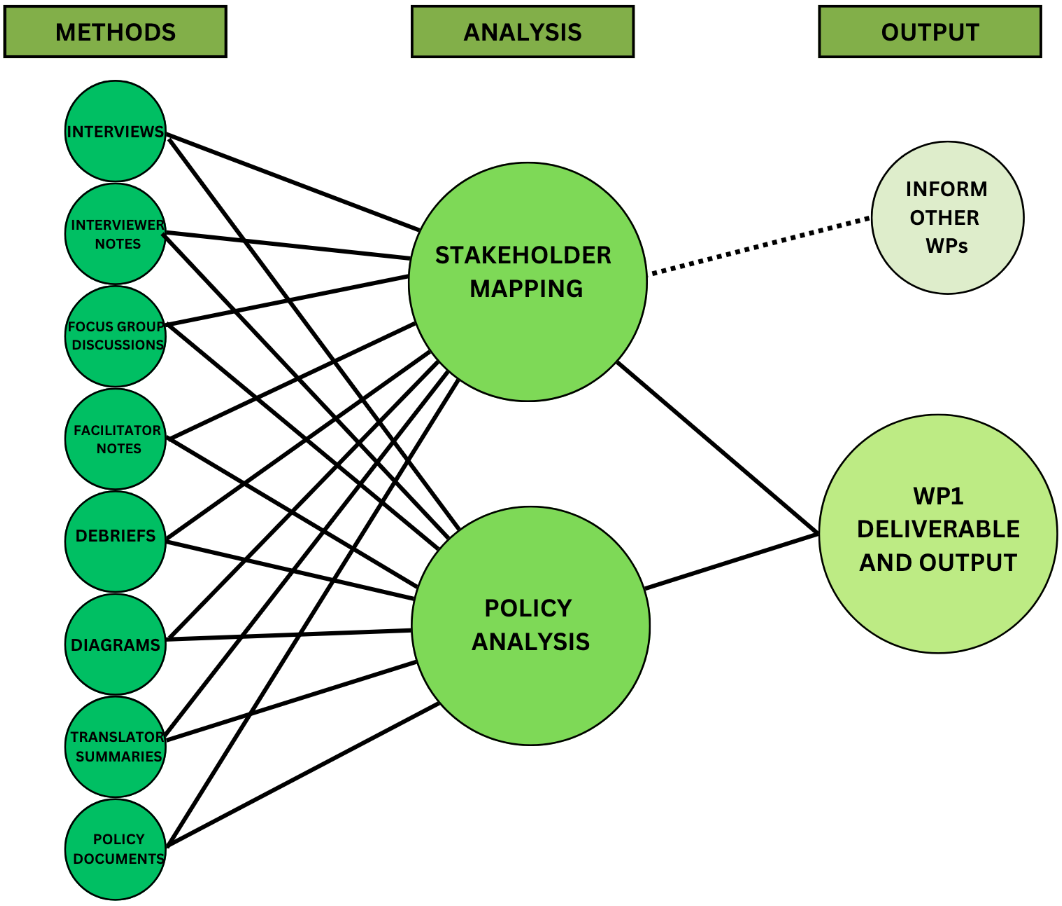

Figure 2: A visualisation of how the data types come together for the WP1 deliverable and output and informs other work packages

1.4 Data Analysis

The underlying methodology for the data analysis was the use of Walt and Gilson's (1994) policy analysis triangle, a systematic approach to consider the different factors influencing policy. The methodology studies together the context, the systemic political, economic, social and cultural factors influencing policy, in addition to content, the substance of the policy and the process, the way in which policies are formulated, implemented and evaluated. However, most emphasis is given to the central concept of actors, the individuals, organisations and the state who have the agency to be initiate, act on and impact policies. These elements of the triangle are synthesized together to understand the policy environment (Zahidie et al 2023). Alongside Walt and Gilson's (1994) policy analysis triangle, a reflexive thematic analysis approach was used for all data sources. This approach was guided by Braun and Clarke (2006) and Campbell et al (2021). A rapid assessment was initially conducted using a deductive analysis approach, this was followed by a full analysis using an inductive analysis approach. Both approaches used the six steps reflexive from Campell et al (2021) data familiarisation, initial code generation, generating (initial) themes, theme defining and naming, theme review and report production, which were adapted for this study. This approach was also used for the policy document analysis. The use of an inductive and deductive analysis allowed us to provide a rapid understanding of the data for the intervention development (WP5) and gaps in the data. Hybrid approaches are useful to use theory driven and theory creating coding, allowing for diverse purposes as part of the data analysis process (Fereday et al., 2006).

1.5 Ethics

WP1 through the Research Ethics Review Committe for Research Involving Human Research Participants at Chulalongkorn University obtained ethics permission (No. 66079). All participants provided their written informed consent to participate in this study. All participation in the study was voluntary and participants were over 18 years old.

4. Results

WP 1 conducted 22 KII and 5 FGDs in Chanthaburi province; 18 KII and 4 FGDs in Chiang Rai province and 7 KIIs 3 FGDs at National Level. In addition, coproduction workshops were run over two days with a total of 35 participants from the national, provincial and NGO level. In addition, 68 policy documents were retrieved, 45 included and 23 excluded. Of those included 5 are international policy documents, 10 are national documents, 1 regional, and 29 provincial policy documents. The majority of the identified study documents were related to disease control and surveillance, particularly those available at the provincial level. While the policy document retrieval process went through an in-depth methodology, a limited number of policies were focused on preparedness and prevention, for which WP1 adapted the methodology from a solely policy document analysis to a mixed data policy analysis, where policy documents are complementing IDI and FDG data.

3.1 Rapid Assessment

WP1 conducted a rapid assessment of Chanthaburi interview and focus group discussion transcripts using a deductive thematic analysis approach. Two researchers firstly reviewed one transcript together and the remaining transcripts were split between the researchers. The deductive approach based on the interview guidelines was used to generate themes which were connected to develop an original framework. In total 12 themes were identified, namely: stakeholder mission, human and animal relationship, multisectoral surveillance, communication, risk, response, learning from COVID-19, challenges, adapting, law, finance and policy. The following sections delves into each theme and highlights the key points emerging from the rapid assessment.

Policy

From the rapid assessment, there appears to be a hierarchical policy influence; local policies are based on district policies, which are then dependent on provincial and national policies and strategies, at times referred to as the 'master plan'. Current policies are created based on well-known zoonotic diseases such as rabies, malaria and dengue. These policies are focused on prevention, surveillance and controlling of epidemics and are put into practice by various actors including members of the department of disease control, livestock, village health volunteers and animal health volunteers. At present there is no general policy focusing on zoonotic risk from human to animal and animal to human transmission generally. However, there are plans on how to address unknown diseases, yet there are conflicting views on the effectiveness of these plans.

Organisation's Mission

Stakeholders interviewed appeared to have a clear vision of their organisation's mission, role and their own responsibility within it. Their roles in relation to pandemic prevention, preparedness and response are seen to be primarily related to surveillance, monitoring, reporting, data evaluation and knowledge transfer with the mission to detect and prevent possibilities from outbreaks in various capacities. However, motivation level varied among stakeholders, often related to workload.

Relationship between humans and animals

Participants refer to various relationships with animals. Domestic animals such as cats and dogs are fondly cared for and treated with affection as pets. Other domestic such as goats, ducks and chicken are reared in and near houses to generate income for the household. Wild animals are also mentioned with multiple references to elephants causing disruption. A wide variety of animals are also a food source, including wild animals, with no fear seen associated with consuming wild meat.

Multisectoral Surveillance

Collaboration and communication are recalled between different sectors and stakeholders such as hospital staff, public health office, disease control committee and district chiefs. A key network of actors was suggested to be village health volunteers, teachers, students, monks, experts, community leaders, and public health officials, who are made aware of 5 key symptoms to look out for amongst community members. The border was indicated to be a key area of surveillance with the public health office working closely with the military at checkpoints.

Communication

An emerging finding was the importance of communicating disease risk to solve problems. Various communication channels are currently being used at the provincial level with clear communication flows existing for known diseases such as rabies and dengue. Communication pathways include in-person meetings face-to-face within provincial departments but also with the community on a regular basis. This was alongside the use of the radio tower to broadcast information, using media including posters and informative signs, and phones in particular the Line app was found to be common. The village leader, village health volunteers or 'main characters' of the village were identified as being key stakeholders to communicate with at the local level to ensure the information is passed on to the villagers. Recommendations included communicating using multiple channels in an accessible local language.

Risk Transmission

The rapid assessment suggests opportunities for zoonotic spillover risk are higher at the borders of Thailand with greater risk coming from the movement of people and animals. The movement of people includes international visitors and migrant workers with greater risk seen from 'others'. Animals particularly dogs are mentioned as frequently moving across borders with a particular fear of illnesses coming from Cambodian dogs. Other areas of risk include injuries from animals such as bites and activities such as cockfighting. A greater risk of disease is indicated to come from domestic animals in comparison to wild animals. Further, raw meat was commonly eaten during festivities and ordinations, however there was growing caution growing due to perceived 'parasites and germs' in raw meat.

Response

The response to exposure from a disease is seen to occur through the local system. For injuries from animals there appears to be a systematic process of action including communication pathways with multiple actors involved. The subdistrict health promoting hospital is often referred to as the first point of contact, followed by the public health office who will inform the livestock department. Village health volunteers will also be notified. If the injury is suspected to be from an animal with rabies, the village headman and kaman will go to the house and lock up the suspected animal.

COVID-19 pandemic events

Key stakeholders during the COVID-19 pandemic appeared to be the public health office, subdistrict hospital, village health volunteers and the village headman. These stakeholders worked together and communicated on a case-to-case basis. The role of the public health office involved controlling the movement of people and organising quarantine facilities. Village health volunteers were considered to have played an important role in the community and village headmen were important to pass on information the villagers. It was thought villagers were more fearful of COVID-19 than local diseases, therefore were more likely to follow the rules and guidelines. However, quarantine rules were breached when villagers ran out of resources. Many lessons were learnt from the COVID-19 pandemic including the need to improve the speed of detection of disease. However, positive behavioral change was also seen amongst community members such as washing hands and wearing masks.

Challenges

A range of challenges were highlighted from a rapid assessment of the interviews. They included a personnel gap with not all local areas having public health officials. Further there was indication that public health officials do not have sufficient power and agency, with the provincial administration being superior. Additionally, the same level of importance was not given to all diseases and there was greater focus on care rather than prevention. Another challenge was budget constraints, particularly for purchasing equipment. Challenges in reaching all members of the community were also shared, particularly houses far away and when individuals are not at home and are away at work. Participants also mentioned difficulties in reaching and vaccinating migrants with no house registration and language barriers at the border. A further challenge at the border is no control of entry and exit of foreign items and animals.

Adapting

Reflections were shared on how to adapt to future policies. A suggestion was to adjust existing policies according to the available budget to the local context and to trial new initiatives at village level before expanding to district level. Further when a new disease is identified, it was recommended a relevant policy should detail steps for multisectoral meetings with clearly defined roles for the stakeholders. Increasing the number of personnel was suggested to carry out tasks effectively and to include the private sector for operations. Additionally, it was suggested a policy should be created for the joint operations of agencies dealing with animals and agencies dealing with people. Areas identified for improvement included the current public relations structure and disease knowledge for villagers particularly those rearing animals.

Law

From the rapid assessment there appeared to be limited mention of the laws in relation to disease transmission and pandemic prevention. It was only mentioned in the context of the illegal trade of wildlife. Further insight into this theme may emerge from the additional interviews being conducted in addition to the policy document analysis.

Finance

Budget constraints were often mentioned particularly at the provincial public health level. Allocation of funds was seen to be dependent on strategic work and determined by higher executives. Budgets were also seen allocated by disease. It was suggested there is an emergency budget, that the local government will use during epidemics primarily for risk areas such as checkpoints at the border. In comparison, there appears to be little budget available for knowledge transfer to the locals such as through media and advertisements. During COVID-19 it was mentioned additional funds were generated by the community with more affluent members helping others in the community. The livestock department appears to have a large budget which comes from the central government and is divided at the provincial level based on work group and level of activity.

Summary

The rapid assessment of the transcripts enabled a preliminary understanding of present zoonotic spillover risk prone areas, activities and behaviours, alongside current pandemic prevention and prepardness efforts particularly at the provincial, district and local level.

Challenges and gaps were also identified indicating areas for improvement and intervention. A full thematic analysis will provide more depth and understanding on these topics.

3.2 Full assessment

WP1 conducted a full inductive thematic analysis. For the inductive analysis, two researchers coded independently three documents. Both researchers worked for the documents, as coder and reviewer. The coder would do the first round of coding and then the reviewer would code above the first coder. This resulted in 450 original codes. Both coders would then together go through those codes to develop themes based on the codes and to merge similar codes and eliminate duplicates. The first round resulted in 99 themes on one level. In a second iteration, the themes were then organised in subcategories, before doing another round of consolidating themes to reduce the number of themes and codes, and therefore complexity. In total, Main themes = 12, Sub-theme 1 = 44, Sub-theme 2=46, Sub-theme 3=14, Sub-theme 4=6, Sub-theme 5 =2, Sub theme 6=1.

In a following stage, two reviewers together looked through one document, at three or more codes for a quote to develop new theoretical themes that were backed up by related theories in diverse disciplines, such as political sciences, anthropology, geography and public health. In this process out of the initial 234 codes, 42 themes were developed. In a final step they were organised in 10 main themes and 23 subthemes. The 10 main themes are: Governance, Pandemic, Case Studies, Stakeholder Mapping, COVID Lessons, Environmental Drivers of Risk, Food Culture, Preparation and Consumption, Health Sector Financing, Communication Channels and Target Audience, Perception and Attitudes.

5. Outlook

Following the four iterations of reflexive thematic analysis, the full assessment of the remaining transcriptions is under way. The timeline is dependent upon receiving the translated transcripts.

Coding Stage	Transcripts Coded	Coders	Codes Generated	Number of Themes Identified	Reflexive Analysis Stage

Stage 1	NGO_01	Coder 1 coded and coder 2 reviewed	450 codes generated	99 themes – one level	Data familiarisation, initial code generation and generating (initial) themes)
	KII_P12	Coder 2 coded and coder 1 reviewed			
	KII_P21	Coder 1 coded and coder 2 reviewed			
	Accelerating One Health in Asia and the Pacific	Coder 2 coded and Coder 1 reviewed			
Stage 2	NGO_01	Coder 1 and coder 2	Same 450 codes	Main themes = 12, sub-theme 1 = 44, sub-theme 2 =46, sub-theme 3 =14, sub-theme 4=6, sub-theme 5=2, sub-theme 6=1	Theme defining and naming
	KII_P12	Coder 2 coded and coder 1 reviewed			
	KII_P21	Coder 1 coded and coder 2 reviewed			
	Accelerating One Health in Asia and the Pacific	Coder 2 coded and coder 1 reviewed			
Stage 3	CT_KII_12	Coder 2 coded and coder 1 reviewed	Using same codes as initial code for transcript = 234 codes	42 themes	Theme Review
Stage 4	CT_KII_12	Coder 2 coded and coder 1 reviewed	Using same codes as initial code for transcript = 234 codes	10 themes, 23 sub-themes	Theme Review

Table 1: *A table describing the different iteration stages of the thematic development*

The process of conducting additional interviews is ongoing and will depend upon availabilities and agreements of additional participants and collaboration with the new Thai partner Khon Kaen University (KKU), who will take over after Chulalongkorn University. The process of retrieving additional policy documents in Thai is expected to continue and finalised with the new Thai partner. The analysis of all data sources will be triangulated and synthesised to inform the final results.

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